

Hierarchical Vietnamese Traffic Sign Detection using YOLO26n and Hierarchical Label Smoothing

Project-Based Learning Report — Milestones R1, R2, R3 (Cumulative)

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This report is cumulative: content from R1 is retained and extended by R2, and R2 is retained and extended by R3, per the milestone requirements. Each subsection below is tagged with its originating milestone in colored brackets — [R1], [R2], or [R3] — so the objective and scope of each addition can be verified independently. Sections with no tag apply across all milestones.

Note: between R2 and R3, the evaluation protocol was revised so that HLS hyperparameter selection happens via cross-validation on the training pool only, with the test set evaluated exactly once per final model after all selection decisions are locked in (the original protocol had used the test set as the validation split during training). All content below reflects the revised protocol; R2's originally-reported interim figure is superseded (see Section 4.6).

1. Introduction

[R1] Problem definition, motivation, application, input/output, scope.

1.1 Problem Definition

Traffic sign detection is the task of localizing and classifying traffic signs within road-scene images. This project addresses Vietnamese traffic sign detection specifically, using the Vietnamese Traffic Signs (VNTS) dataset, which contains 3,216 images and 8,334 annotated instances across 52 sign classes governed by the QCVN 41 national road signage regulation.

1.2 Motivation and Why This Topic Was Chosen

Existing Vietnamese traffic sign datasets and models treat all sign classes as independent (“flat-label”), ignoring the hierarchical regulatory structure defined by QCVN 41, under which every sign belongs to one of five broad categories: Prohibitory, Warning, Mandatory, Information, and Supplementary. This structure is a natural and unused source of supervision.

The dataset also exhibits severe class imbalance (instance counts ranging from 3 to 1,071 per class), which is representative of real deployment conditions but makes rare-class detection difficult under standard training.

This project investigates whether exploiting the QCVN 41 hierarchy through the training loss can improve detection accuracy, particularly for underrepresented sign classes, and separately whether an architecture-level small-object detection enhancement (a P2 detection head) provides an independent benefit — questions not previously studied together for Vietnamese traffic signs.

1.3 Application

- Advanced Driver Assistance Systems (ADAS) — real-time sign recognition for driver alerts.
- Autonomous vehicle perception stacks operating in Vietnamese road environments.
- Automated road-infrastructure auditing (e.g. verifying signage compliance with QCVN 41).

1.4 Input, Output, and Scope

Item	Definition
Input	RGB road-scene image (640×640 after resizing)
Output	Set of bounding boxes, each with a predicted class among 52 QCVN 41 sign identifiers
Scope	Detection only (not tracking or depth estimation); single-frame, not video
Out of scope	Real-time embedded deployment, OCR of sign text (e.g. speed values), multi-camera fusion

1.5 Planned Methods

The project trains and compares three models, isolating two independent design questions against the same flat-label baseline:

- **Model B (baseline):** unmodified YOLO26n, standard training procedure.
- **Model A (loss-level intervention):** architecturally identical to Model B, but trained with Hierarchical Label Smoothing (HLS), which injects the QCVN 41 taxonomy into the classification loss target distribution without modifying the model architecture. HLS's hyperparameters (target-gap width, classification loss weight) are selected via cross-validation on the training pool, not fixed a priori and not selected against the test set.
- **Model P2 (architecture-level intervention):** YOLO26n augmented with an additional P2 detection head for higher-resolution small-object detection, trained with the same flat loss target as Model B, isolating architecture as the sole variable in that comparison.

This differs from prior hierarchical detection work (see Related Works), which typically requires additional detection-head branches or multi-stage cascades for the hierarchical signal specifically; here the hierarchical (HLS) and architectural (P2) interventions are tested independently of one another, each against the same control.

1.6 Evaluation Metrics

- Precision, Recall (macro-averaged across classes)
- mAP@0.5 and mAP@0.5:0.95 (standard COCO-style detection metrics)
- Per-class AP50, stratified by frequency tier (Frequent / Medium / Rare)
- Wilcoxon signed-rank test on paired per-class AP50 scores (Model A vs. Model B, Model P2 vs. Model B), computed once on the held-out test set after all hyperparameter and checkpoint decisions are finalized

1.7 Data Source

The VNTS dataset is a pre-existing, publicly available dataset (not self-collected), obtained via Kaggle. Annotations are provided in YOLO format (class ID, normalized bounding box). No manual re-annotation was performed; the QCVN 41 category mapping was derived programmatically from a deterministic lookup table built from the QCVN 41 regulation text.

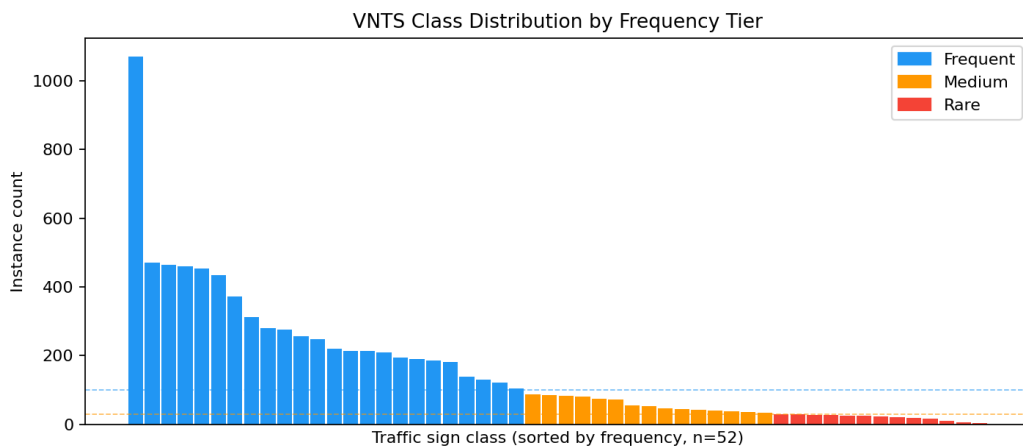


Figure 1. VNTS class distribution across 52 sign classes, color-coded by frequency tier (blue = Frequent ≥ 100 instances, orange = Medium 30–99, red = Rare < 30).

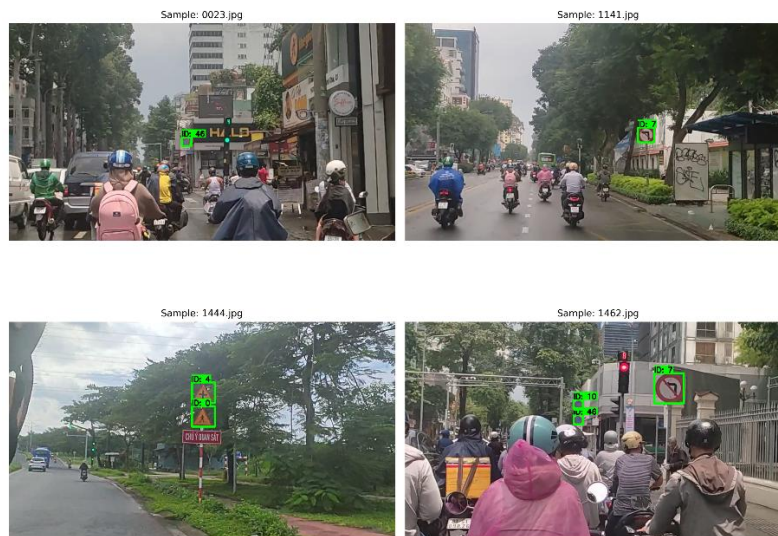


Figure 2. Sample images.

1.8 Pipeline Overview

The end-to-end experimental pipeline runs from raw data through taxonomy mapping, HLS hyperparameter selection, final training of all three models, and one-time test-set evaluation, as summarized below.

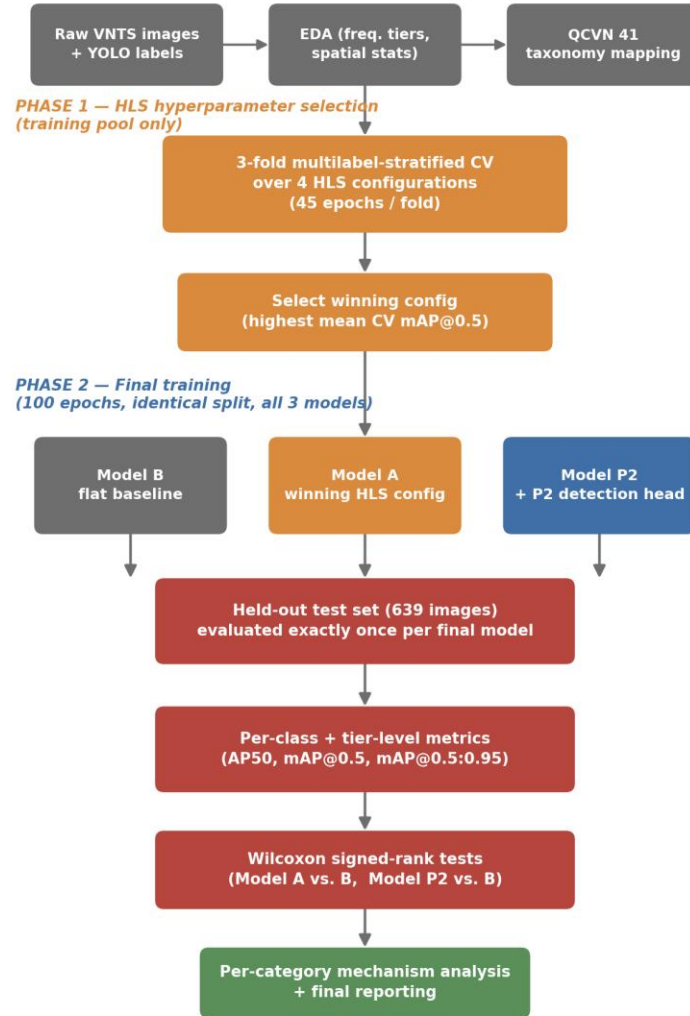


Figure 3. Training and evaluation pipeline, from raw data to final reporting.

2. Related Works

[R2] Related works (mandatory, continued from R1).

Lu et al. proposed a two-stage hierarchical traffic sign recognition framework that first predicts broad sign shape categories before identifying specific sign classes, demonstrating that coarse-to-fine prediction improves accuracy over flat-label approaches. This project similarly exploits a coarse-to-fine structure, but implements it through the loss function of a single-stage detector rather than a sequential two-stage pipeline, preserving inference speed.

Zhang et al. incorporated hierarchical semantic relationships into a traffic sign detection framework via structured category-level feature learning. This project tests the same underlying principle at the loss level rather than the architecture level.

Usmani et al. proposed a hierarchical YOLOv8-based detector with real-time text recognition for UAE traffic signs. Their hierarchy is defined by sign shape and text content for a UAE-specific sign system, and their study does not investigate the sensitivity of hierarchical supervision to loss-weighting hyperparameters — a factor this project identifies as critical (Section 5.2).

Tsenkova et al. introduced hYOLO, a hierarchical extension of YOLOv8 with a modified hierarchical loss function. Their reported finding that increasing classification loss weight improves hierarchical YOLO performance motivated the classification-weight ablation conducted in this project; the same qualitative pattern (higher weight helps within the HLS

family) is observed here, though the improvement does not carry through to a significant gain over a flat baseline (Section 5.4).

Zhu et al. introduced TPH-YOLOv5, which adds a dedicated high-resolution detection head (“P2”, stride 4) sourced from an early backbone stage, extending detectable object size down to roughly 4×4 pixels, specifically to improve small- and tiny-object recall on drone-captured imagery. Since VNTS sign instances are uniformly small relative to image resolution (Section 3.4), this project tests whether the same architectural enhancement transfers to traffic sign detection, independent of the loss-level HLS intervention.

Ertler et al. introduced the Mapillary Traffic Sign Dataset, one of the largest traffic sign benchmarks available at global scale. It does not include a taxonomy aligned with Vietnamese QCVN 41 regulations, motivating the use of VNTS despite its smaller scale.

Sapkota et al. detail the architectural enhancements of YOLO26, including NMS-free end-to-end inference and the MuSGD optimizer, which form the base architecture used in this project.

Genermont et al. investigate hierarchical novelty detection for traffic sign recognition, motivating the Hierarchical Cosine Loss direction identified as future work in this project (Section 6.3).

3. Data: Exploratory Data Analysis and Preprocessing

[R2] Data completion — analysis, cleaning, visualization.

3.1 Dataset Overview

Property	Value
Total images	3,216
Total annotated instances	8,334
Sign classes	52
Train / test split	2,552 / 639
Average objects per image	2.59

3.2 Class Imbalance and Frequency Tiers

Instance counts per class range from 3 to 1,071, forming a severe long-tail distribution. Classes were stratified into three tiers using the 25th and 75th percentiles of the instance-count distribution as cutoffs:

Tier	Threshold	Classes	Instances
Frequent	≥ 100	24	7,201
Medium	30–99	15	873
Rare	< 30	13	260

3.3 QCVN 41 Hierarchical Taxonomy

All 52 classes were mapped to five QCVN 41 Level-1 categories via a deterministic lookup table. Two label inconsistencies in the raw dataset were identified and resolved during this process: a class labeled “B.8a” was identified as P.108a (a Prohibitory sign) based on visual inspection, and a class labeled “Camera” (referring to speed-camera warning signs, not a standard QCVN 41 code) was assigned to the Information category as a practical decision.

Category	Classes	Instances	% of Total
Prohibitory	23	5,147	61.8%
Warning	18	1,727	20.7%

Mandatory	4	798	9.6%
Information	4	472	5.7%
Supplementary	3	190	2.3%

3.4 Spatial Characteristics

Median bounding box area remains under 0.25% of image resolution across all three frequency tiers (Frequent 0.14%, Medium 0.24%, Rare 0.19%), confirming that object scale does not confound tier-based performance comparisons — signs are uniformly small regardless of class frequency.

This uniform smallness is the motivation for testing the P2 architectural enhancement (Section 4.4): if a meaningful fraction of instances fall near or below the effective resolution of YOLO26n's standard highest-resolution head (P3, stride 8), an additional stride-4 head could plausibly help regardless of class frequency.

4. Model Architecture and Methodology

[R2] Model architecture details.

4.1 Base Architecture

All three models build on YOLO26n: a CSP-based convolutional backbone, PAN-FPN multi-scale neck, and an anchor-free, NMS-free end-to-end detection head.

Models A and B use the architecture unmodified (2,384,976 parameters, 5.2 GFLOPs) and are architecturally identical to one another — the only difference between them is the classification loss target (Sections 4.2–4.3).

Model P2 adds a fourth, higher-resolution detection head (Section 4.4), increasing parameter count to 2,429,832 (+1.9%) and compute to 6.8 GFLOPs (+30.8%).

4.2 Model B — Flat-Label Baseline

Model B uses YOLO26n's standard training procedure without modification. In practice, the classification target at a positive anchor is not a literal one-hot vector: YOLO26n's Task-Aligned Assigner (TAL) scales the target for the assigned ground-truth class by a box-quality metric (prediction confidence combined with IoU), so that poorly localized predictions receive a lower target even for the correct class.

Critically, all other 51 classes remain exactly zero — TAL performs quality-scaled confidence adjustment for a single class, not cross-class redistribution. All misclassified classes are therefore still penalized identically regardless of semantic proximity, since none receive any target probability.

4.3 Model A — Hierarchical Label Smoothing (HLS)

Let $S(c)$ denote the set of sign identifiers sharing class c 's QCVN 41 Level-1 category, including c itself. The HLS target redistributes probability mass across three groups:

- the correct class receives $1 - \alpha - \beta$;
- each of the $|S(c)| - 1$ sibling classes within the same category receives $\beta / (|S(c)| - 1)$;
- each of the remaining classes outside the category receives $\alpha / (N - |S(c)|)$, where $N = 52$.

This smoothed target replaces the TAL-scaled target at positive anchors inside the standard binary cross-entropy classification loss; the bounding box regression and distribution focal loss components are unchanged. HLS is applied identically to both the one-to-many and one-to-one assignment heads of YOLO26n's end-to-end loss.

Note that this substitution fixes the correct-class target at $1 - \alpha - \beta$ regardless of box localization quality, discarding the TAL quality-scaling signal that Model B retains — this trade-off is revisited in the Limitations section.

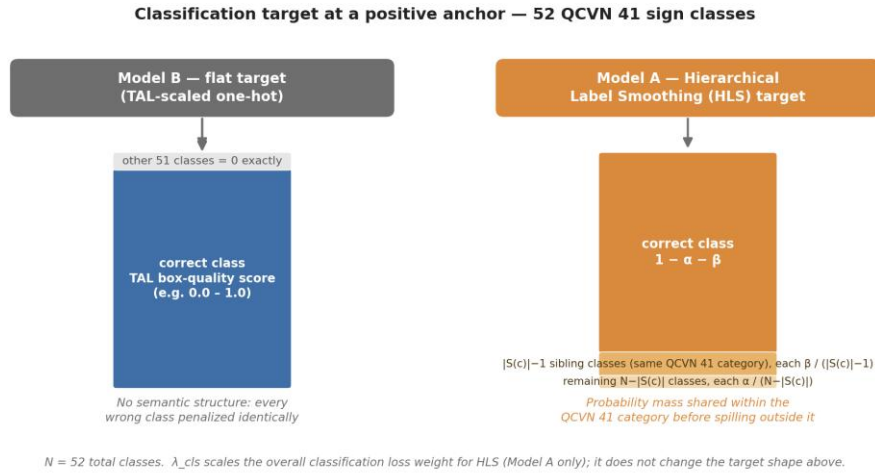


Figure 4. How Model A (HLS) redistributes classification-target probability mass, compared with Model B's flat one-hot target.

As a concrete illustration, consider a class c in the Warning category (18 classes, so $|S(c)| = 18$), under Model A's final hyperparameters ($\alpha = 0.05$, $\beta = 0.10$, $N = 52$): the correct class receives $1 - 0.05 - 0.10 = 0.85$; each of the other 17 Warning-category classes receives $0.10 / 17 \approx 0.0059$; each of the remaining 34 classes outside Warning receives $0.05 / 34 \approx 0.0015$. These three groups sum to exactly 1.0.

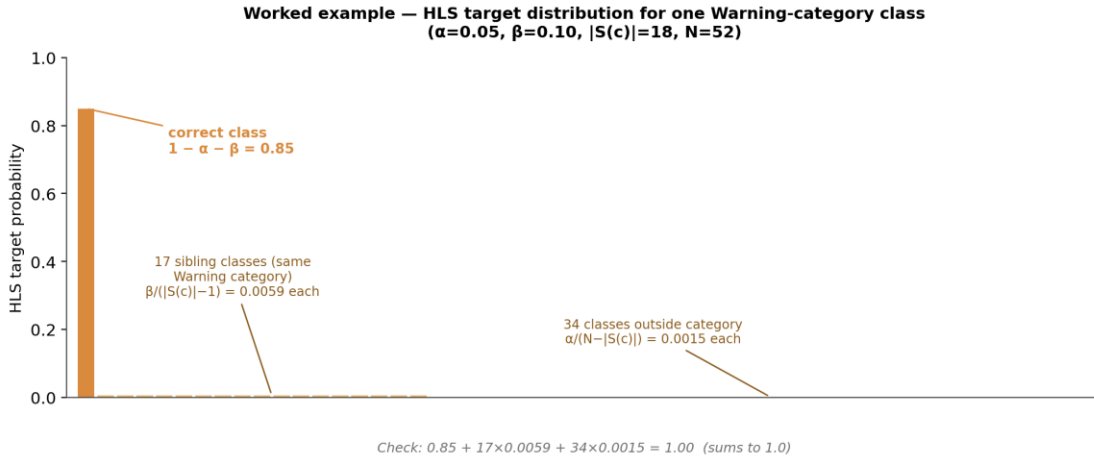


Figure 5. Worked example — HLS target distribution across all 52 classes for one Warning-category class.

The (α , β , λ_{cls}) hyperparameters used for Model A's final training were not fixed in advance — they are the winning configuration selected via the cross-validation ablation described in Section 4.5 and reported in Section 5.2.

4.4 Model P2 — Architectural Small-Object Enhancement

Model P2 adds a dedicated high-resolution detection head at stride 4 (“P2”), following the extra-head design used in TPH-YOLOv5 for small- and tiny-object detection. P2 features are sourced from an earlier backbone stage than YOLO26n's default earliest head (P3, stride 8) and fused with upsampled P3 features in the neck, extending the range of reliably detectable object sizes downward.

Model P2 is initialized from the pretrained yolo26n.pt checkpoint wherever layer shapes match; the new P2-specific layers are randomly initialized, since no pretrained weights exist for them.

Model P2 uses the same flat (non-hierarchical) classification target as Model B — architecture is the sole independent variable in the Model P2 vs. Model B comparison, isolated from the loss-level HLS intervention.

4.5 Training Configuration

Training proceeds in two phases, structured to keep the 639-image held-out test set completely unused until every hyperparameter and checkpoint decision has already been made.

Phase 1 — HLS hyperparameter selection (Model A only). Four HLS configurations (varying α , β , and classification loss weight λ_{cls}) are evaluated via 3-fold multilabel-stratified cross-validation, restricted entirely to the 2,552-image training pool.

Each fold-configuration combination trains for 45 epochs — a budget set by Kaggle's per-session runtime limit (12 hours) and weekly GPU quota (30 hours): the ablation requires 12 fold-configuration runs (4 configurations $\times$ 3 folds), which could not be extended to a longer per-run schedule without exceeding the weekly quota; the configuration with the highest mean cross-validation mAP@0.5 (Section 5.2) is selected as Model A's final configuration. The flat baseline is not part of this ablation — it is evaluated separately as Model B — since the ablation's purpose is choosing among HLS variants, not deciding whether HLS beats no-HLS.

Phase 2 — Final training (all three models). The winning HLS configuration (Model A), the flat baseline (Model B), and the P2-head baseline (Model P2) are each trained for 100 epochs on an identical split: approximately 90% of the training pool for training, and a stratified internal validation carve-out (~10%) used only for best-checkpoint selection during training.

All three final models are trained and evaluated on exactly the same split, generated once and reused across all three, so no model has a data advantage relative to the others.

Test evaluation. The 639-image test set is evaluated exactly once per final model (Model A, Model B, Model P2), after training and checkpoint selection are complete. No configuration, hyperparameter, or checkpoint decision anywhere in the pipeline is made by looking at test-set performance.

Parameter	Value
Base weights	yolo26n.pt (COCO-pretrained)
Input resolution	640 $\times$ 640
CV ablation epochs (per fold)	45
CV folds	3 (multilabel-stratified)
Final training epochs	100
Internal validation carve-out	~10% of training pool
Batch size	32
Optimizer	AdamW (auto-selected)
Augmentation	Mosaic, HSV jitter, horizontal flip
Random seed	0 (fixed across all runs)
Hardware	NVIDIA Tesla T4 (Kaggle)
Experiment tracking	Weights & Biases (wandb)

Because HLS modifies only the classification target via a single lookup into a precomputed 52 $\times$ 52 smoothing matrix per batch, it adds negligible training overhead and has no effect on inference: at inference time, Model A and Model B execute an identical forward pass through an identical architecture, since the loss function is not part of the inference computation graph.

Inference cost is therefore identical between Models A and B by construction, not merely by measurement. Model P2's inference cost differs from both by construction, due to its additional detection head (Section 4.1).

4.6 Note on R2's Interim Result

R2 reported an interim baseline figure (mAP@0.5 = 0.9487, mAP@0.5:0.95 = 0.7171) under an earlier evaluation protocol that used the test set as the validation split during training; that protocol was revised prior to R3, so this figure is not directly

comparable to the results below and is superseded by Section 5.3, which reports Model B's performance under the revised protocol.

5. Final Results and Comparative Evaluation

[R3] Final results — minimum 3 models (3 models trained and compared).

5.1 Models Trained

Three final models are compared, each evaluated on the held-out test set exactly once:

Model	Architecture	Loss target	Selection
Model B	YOLO26n (unmodified)	Flat (TAL-scaled one-hot)	Fixed, no tuning
Model A	YOLO26n (unmodified)	HLS: $\alpha=0.05$, $\beta=0.10$, $\lambda_{\text{cls}}=1.5$	Winner of 4-way CV ablation (Section 5.2)
Model P2	YOLO26n + P2 head	Flat (TAL-scaled one-hot)	Fixed, no tuning

An additional four HLS configurations were trained during the Phase 1 cross-validation ablation to select Model A's configuration; those are reported in Section 5.2 but are not part of the final three-model comparison.

5.2 HLS Hyperparameter Selection (Cross-Validation)

All four configurations were evaluated identically: 3-fold multilabel-stratified cross-validation, 45 epochs per fold, restricted to the training pool. Values below are mean $\pm$ standard deviation across folds.

Config	α	β	λ_{cls}	Global	Freq	Med	Rare
HLS-1	0.05	0.10	0.5	0.579 $\pm$ 0.005	0.752 $\pm$ 0.009	0.551 $\pm$ 0.030	0.285 $\pm$ 0.013
HLS-2_wide	0.01	0.20	0.5	0.527 $\pm$ 0.009	0.732 $\pm$ 0.005	0.482 $\pm$ 0.027	0.193 $\pm$ 0.020
HLS-3_hcls ★	0.05	0.10	1.5	0.687 $\pm$ 0.008	0.808 $\pm$ 0.006	0.733 $\pm$ 0.008	0.403 $\pm$ 0.023
HLS-4_combo	0.01	0.20	1.5	0.656 $\pm$ 0.005	0.796 $\pm$ 0.002	0.672 $\pm$ 0.015	0.370 $\pm$ 0.017

Within the HLS family, classification loss weight (λ_{cls}) is the dominant driver of performance: both $\lambda_{\text{cls}}=1.5$ configurations (HLS-3_hcls, HLS-4_combo) substantially outperform both $\lambda_{\text{cls}}=0.5$ configurations (HLS-1, HLS-2_wide) at every tier, most dramatically in the Rare tier (0.37–0.40 vs. 0.19–0.29). Widening the target gap alone, without a correspondingly higher classification weight (HLS-2_wide vs. HLS-1), degrades performance rather than helping.

HLS-3_hcls ($\alpha=0.05$, $\beta=0.10$, $\lambda_{\text{cls}}=1.5$) achieved the highest mean cross-validation mAP@0.5 at every tier and was selected as Model A's final configuration. This λ_{cls} effect is both large and low-variance — fold-level standard deviations stay within roughly 0.005–0.03 across all four configurations — so it is a real, reproducible effect within the HLS family, not a fold-sampling artifact.

Robustness within the family is a separate question from whether the winning configuration beats the flat baseline, however; Section 5.3 shows it does not.

Methodology note: Because CV ablation used a shorter epoch budget (45) than final training (100), constrained by Kaggle's 12-hour per-session limit and 30-hour weekly GPU quota, this ranking is not independently verified at 100 epochs for the non-selected configurations — see Limitations (Section 6.2).

5.3 Final Test-Set Results

Each model evaluated once on the 639-image held-out test set, after training and checkpoint selection were complete.

Metric	Model B	Model A	Model P2
Precision	0.8337	0.8295	0.8455
Recall	0.8796	0.8689	0.6708

mAP@0.5	0.9257	0.9131	0.8335
mAP@0.5:0.95	0.6945	0.6690	0.6173
Tier	Model B mAP@0.5	Model A mAP@0.5	Model P2 mAP@0.5
Frequent	0.9536	0.9556	0.8839
Medium	0.9336	0.9387	0.8523
Rare	0.8649	0.8053	0.7188

Model A is close to Model B at every tier: it edges ahead of Model B on Frequent and Medium by less than one percentage point, but trails on both global metrics and, most notably, on the Rare tier (0.8053 vs. 0.8649) — the same tier where the CV ablation (Section 5.2) showed λ_{cls} having its largest effect.

The honest reading is therefore not "HLS does nothing": within the HLS family, λ_{cls} matters a great deal and reproducibly. But even the best-tuned configuration, trained and tested once under the corrected protocol, does not clear the bar the flat baseline already sets.

Model P2 underperforms Model B substantially at every tier and metric, with the largest gap in Recall (0.6708 vs. 0.8796).

5.4 Statistical Comparison vs. Baseline

Wilcoxon signed-rank test on paired per-class AP50 scores, computed once on the test-set results above.

Tier	n	Model A vs. B (p)	Model P2 vs. B (p)
Global	52	0.5539	3.4×10^{-7}
Frequent	24	0.8774	3.6×10^{-7}
Medium	15	0.8139	0.0131
Rare	13	0.2439	0.0645

Model A shows no statistically significant difference from Model B at any tier (all $p > 0.05$): there is no evidence that the CV-selected HLS configuration improves on the flat baseline.

Model P2, by contrast, differs significantly from Model B at the Global, Frequent, and Medium tiers ($p < 0.05$); the Rare tier approaches significance ($p=0.0645$) in the same negative direction. Model P2's difference from baseline is a clear, statistically supported negative result, not a null one.

5.5 Per-Category Analysis (Model A vs. Model B)

Per-class AP50 deltas (Model A minus Model B) were grouped by QCVN 41 category, to test whether category size predicts which classes benefit from HLS, as hypothesized in R2.

Category	Classes	Mean AP50 delta
Supplementary	3	-0.1527
Mandatory	4	+0.0281
Information	4	+0.0344
Warning	18	-0.0093
Prohibitory	23	-0.0120

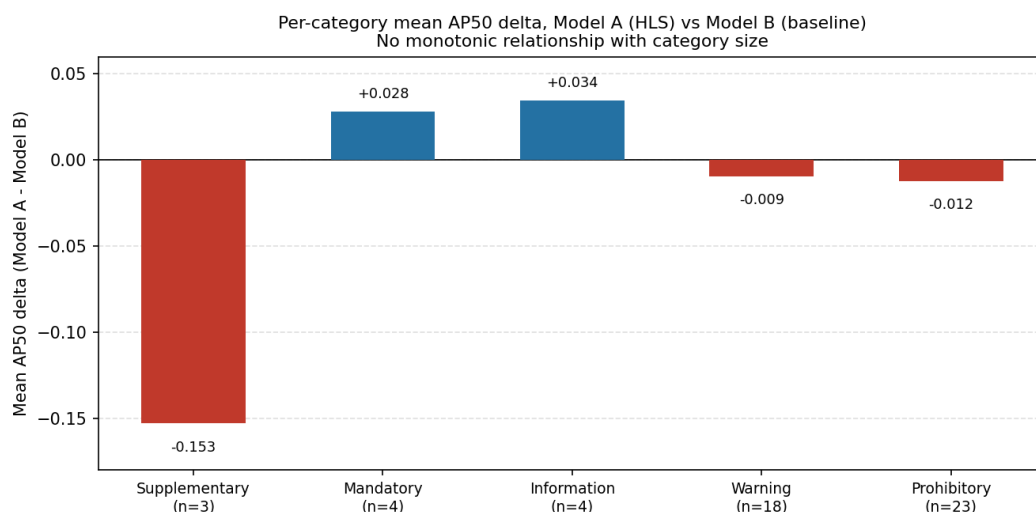


Figure 6. Mean AP50 delta (Model A – Model B) by QCVN 41 category. No monotonic relationship with category size.

The category-size hypothesis from R2 does not survive contact with the taxonomy's sample sizes. Supplementary has only 3 classes and 190 instances total, so a single confused or misclassified class can swing its category-level mean substantially; against that, its large negative delta (-0.153) is at least as consistent with sampling noise from a 3-class category as with a genuine categorical effect, and the same caveat weakens confidence in the small positive deltas for Mandatory and Information (4 classes each).

Rather than concluding that category size does not matter, the honest conclusion is that this taxonomy does not contain enough classes per category to distinguish a real category-size effect from noise — Supplementary, Mandatory, and Information are all too small to support a stable per-category estimate. The two largest categories (Warning, 18 classes; Prohibitory, 23 classes) are the only ones large enough for their mean delta to be a reasonably stable estimate, and both show small losses rather than gains.

A Pearson correlation between raw per-class training-instance count and AP50 delta was also negligible and non-significant ($r=0.0737$, $p=0.6037$), consistent with class rarity not predicting HLS's effect either — see Limitations.

5.6 Architecture-Only Comparison: Model P2

Model P2's failure has a specific signature, not a diffuse one. Precision (0.8455) is the highest of all three models — above both Model A and Model B — while Recall collapses to 0.6708, and $mAP@0.5$ (a threshold-independent metric that integrates over the full precision-recall curve, not just one operating point) also drops clearly, from 0.9257 (Model B) to 0.8335.

This combination rules out the simplest interpretation, that Model P2 got worse at detecting and classifying signs across the board: a model with a genuine localization or classification failure would not simultaneously post the highest precision of the three. Instead, the pattern points to poorly calibrated confidence scores from the new, untrained P2 branch — many of the model's detections are plausibly correct but sit just below whatever confidence threshold Recall is measured at, and the $mAP@0.5$ drop shows this is not an artifact of one threshold choice.

The most likely cause of that miscalibration is that Model P2 introduces new, randomly-initialized parameters (the P2 head and its fusion path) that receive no benefit from COCO pretraining and must learn confidence calibration from scratch within a 2,552-image training pool and a fixed 100-epoch budget — unlike Models A and B, whose parameters are entirely inherited from the pretrained checkpoint and do not face the same cold start.

A secondary, non-exclusive possibility is that VNTS signs are already adequately resolved by YOLO26n's default P3 (stride 8) head — Section 3.4 established only that instances are uniformly small, not that they fall below P3's resolution floor — in which case the added P2 head introduces competing anchors without addressing a real resolution bottleneck.

Both explanations point to the same practical conclusion: this project's data and training budget do not support the P2 architectural change as implemented, and the precision/recall/ $mAP50$ pattern specifically implicates confidence calibration rather than a wholesale inability to localize or classify signs.

5.7 End-to-End System Pipeline

[R3] Full pipeline, raw input to final output.

The figure below traces a single image through the deployed model, from raw input to the final scored detections; offline evaluation (step 6) applies only during experimentation, not at deployment.

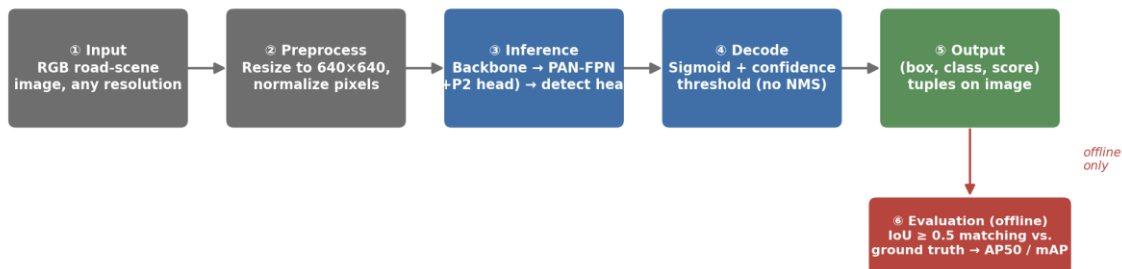


Figure 7. End-to-end inference pipeline, raw image to scored detections.

- 1. Input: raw RGB road-scene image, any resolution.
- 2. Preprocessing: resize to 640×640, normalize pixel values.
- 3. Inference: single forward pass through YOLO26n backbone → PAN-FPN neck (plus P2 head for Model P2) → end-to-end detection head.
- 4. Output decoding: sigmoid activation on class logits, confidence thresholding (no NMS post-processing required — YOLO26n is NMS-free).
- 5. Final output: list of (bounding box, class label, confidence score) tuples overlaid on the input image.
- 6. Evaluation (offline only): predictions compared against ground truth via $\text{IoU} \geq 0.5$ matching, aggregated into per-class AP50, tier-level mAP, and Wilcoxon significance tests.

6. Discussion, Limitations, and Future Work

6.1 Discussion

The central finding of this project is that HLS's apparent benefit does not hold up under a properly separated evaluation protocol: R2's interim result and the original R3 draft both reported a statistically-flavored positive effect for HLS that did not survive a later revision to the evaluation pipeline.

Under the corrected protocol — HLS hyperparameters selected via cross-validation on the training pool only, test set evaluated exactly once — the best HLS configuration (HLS-3_hicls) is statistically indistinguishable from a flat-label baseline at every frequency tier, and on raw numbers actually trails the baseline on both global metrics and the Rare tier, only edging ahead on Frequent and Medium by less than one percentage point.

This does not mean HLS hyperparameters are unimportant — quite the opposite: within the HLS family, classification loss weight (λ_{cls}) drives a large, low-variance effect across CV folds, roughly doubling Rare-tier performance when moved from 0.5 to 1.5 (Section 5.2). The honest summary is therefore not "HLS does nothing," but that HLS's own hyperparameters matter a great deal, and even once they are optimized, the result does not clear the bar a plain baseline already sets.

The category-size mechanism hypothesized in R2 to explain HLS's effect does not survive contact with the taxonomy's sample sizes (Section 5.5): several categories, including Supplementary, contain too few classes for their mean AP50 delta to be distinguished from sampling noise, so the corrected analysis can neither confirm nor rule out a genuine category-size effect — it can only say that R2's specific pattern is not reliable evidence for one.

Separately, the P2 architectural enhancement produces a clear, statistically significant negative result — the most confident finding in this project. Its signature is specific rather than diffuse: the highest precision of the three models, paired with a recall collapse and a corresponding drop in the threshold-independent $\text{mAP}@0.5$, points to poorly calibrated confidence from

the new, untrained P2 branch, most plausibly because those parameters must learn calibration from scratch within a small-dataset, fixed-epoch budget, rather than a fundamental mismatch between P2 heads and traffic sign detection.

6.2 Limitations

- HLS replaces YOLO26n's Task-Aligned Assigner box-quality scaling with a fixed hierarchical target distribution at positive anchors, so Model A's correct-class target no longer reflects localization quality the way Model B's does — a secondary difference beyond the intended hierarchical redistribution, and a candidate confound for future refinement.
- The cross-validation ablation (Section 5.2) used a 45-epoch budget, constrained by Kaggle's 12-hour per-session runtime limit and 30-hour weekly GPU quota across the ablation's 12 fold-configuration runs, while final training used 100 epochs; the relative ranking among the four HLS configurations is not independently verified at the longer schedule beyond the single winning configuration that was carried forward.
- The internal validation carve-out used for final-model checkpoint selection is small (~10% of the training pool, roughly 255 images), which may introduce checkpoint-selection noise independent of the underlying model quality.
- A single random seed was used throughout final training; variance is only characterized across CV folds during the ablation phase, not across independent training seeds for the final models.
- The Supplementary category (3 classes, 190 instances) and the Rare tier (13 classes, 260 instances) produce high-variance per-class AP50 estimates regardless of model configuration, limiting confidence in category- and tier-specific conclusions drawn from them.
- Model P2's negative result was not decomposed into its two candidate causes (undertrained new parameters vs. genuine architecture–task mismatch); Section 6.3 proposes experiments to distinguish them.
- The VNTS dataset contains only 3,216 images; the interaction between either intervention and dataset scale remains untested.

6.3 Future Work

- Repeat Model P2 training with a longer schedule or a staged warm-up (e.g. freezing pretrained layers while the P2 head trains first) to test whether its negative result is a convergence artifact rather than a genuine mismatch with sign object sizes.
- Empirically compare the VNTS bounding-box size distribution against YOLO26n's P2/P3 resolution boundary directly, rather than inferring smallness only from percentage-of-image-area statistics, to determine whether P2's added resolution addresses a real bottleneck for this dataset.
- Repeat the HLS cross-validation ablation with multiple random seeds per configuration to separate seed variance from fold variance, and verify the winning configuration's ranking at the full 100-epoch budget.
- Since neither raw instance count ($r=0.07$) nor QCVN 41 category size predicts HLS's per-class effect, investigate alternative structuring principles (e.g. visual or shape similarity) for a hierarchical target distribution.
- Evaluate on a larger Vietnamese traffic sign dataset to increase statistical power in the Rare tier and Supplementary category.
- Investigate Hierarchical Cosine Loss (Genermont et al.) as an alternative hierarchical formulation; HLS's null result here does not rule out other hierarchical loss designs.

7. Conclusion

This project constructed a QCVN 41-aligned hierarchical taxonomy for Vietnamese traffic sign detection and conducted a controlled empirical investigation of two independent interventions against a flat-label YOLO26n baseline: a loss-level hierarchical label smoothing scheme (Model A) and an architecture-level small-object detection head (Model P2). HLS's hyperparameters were selected via cross-validation on the training pool alone, and the test set was evaluated exactly once per final model — a protocol adopted after an earlier pipeline version was found to use the test set as validation during training.

Under this corrected protocol, Model A's CV-selected configuration ($\alpha=0.05$, $\beta=0.10$, $\lambda_{\text{cls}}=1.5$) is not statistically distinguishable from the flat baseline at any frequency tier and trails it on raw Global and Rare-tier numbers, even though λ_{cls} has a large, reproducible effect within the HLS family itself (Section 5.2) — HLS's hyperparameters matter, but tuning them does not close the gap to a flat baseline.

The category-size mechanism that appeared to explain HLS's effect under the original evaluation protocol can be neither confirmed nor ruled out under the revised per-category analysis, since several categories contain too few classes to separate a genuine effect from sampling noise. Model P2 produces a clear, statistically significant negative result at every tier; its precision/recall/mAP50 signature specifically implicates poorly calibrated confidence in the new, undertrained P2 branch rather than a wholesale loss of detection or classification ability.

Taken together, the project's central contribution is the finding that HLS's apparent benefit does not survive a properly separated evaluation protocol: a plausible-sounding hierarchical loss modification does not outperform a well-tuned flat baseline once hyperparameter selection and final test evaluation are kept independent.

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